

Skill Issue

Decomposing Procedural Knowledge Injection for LLM-Agent Network Forensics

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Agent Skills now specialize LLM agents, yet no study tests whether they make a packet-forensics agent competent, or which part does the work

- Skills package procedural knowledge (workflow, tool recipes, output contracts) injected into a generalist model at inference time.
- Cross-domain evaluation reports cybersecurity as the largest-gain domain, but treats the skill as one atomic block.
- **Open question 1: does a skill make a tshark-driving agent a competent analyst, measured against vetted ground truth?**
- **Open question 2: which component of a skill produces any gain?**

We hold the model and toolbox fixed and vary only the injected skill across six arms, isolating each component by ablation

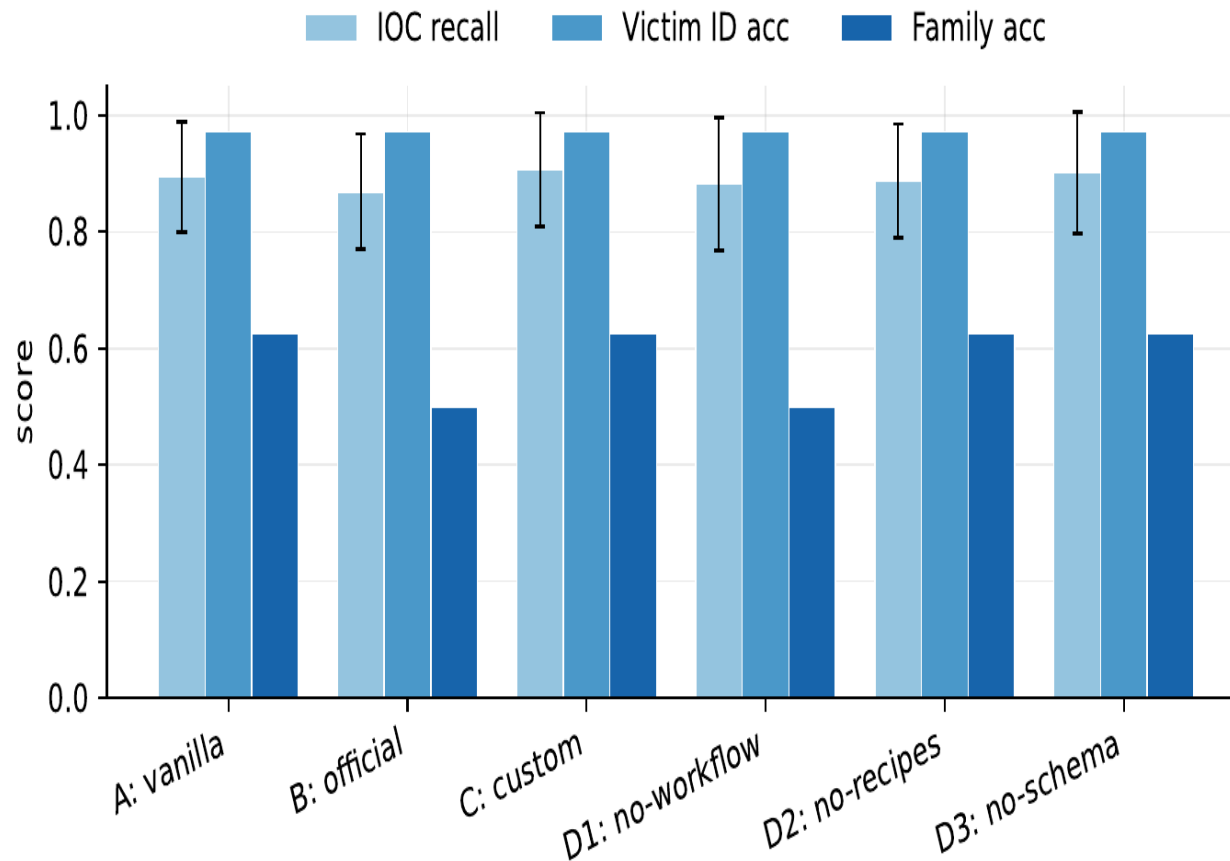
Arm	Injected skill	Components
A	none (vanilla)	baseline
B	community Wireshark skill	external
C	custom IOC-forensics skill	workflow + recipes + schema
D1	C minus workflow	recipes + schema
D2	C minus recipes	workflow + schema
D3	C minus schema	workflow + recipes

- Same neutral task: investigate the evidence, write an incident report.
- Ablations remove one marked section verbatim, so each differs from C by one component only.
- Runtime: Claude Code CLI, read-only tools, web and file-transfer denied.

Validating every indicator against packets exposed wrong C2 IPs in three of nine published answer keys

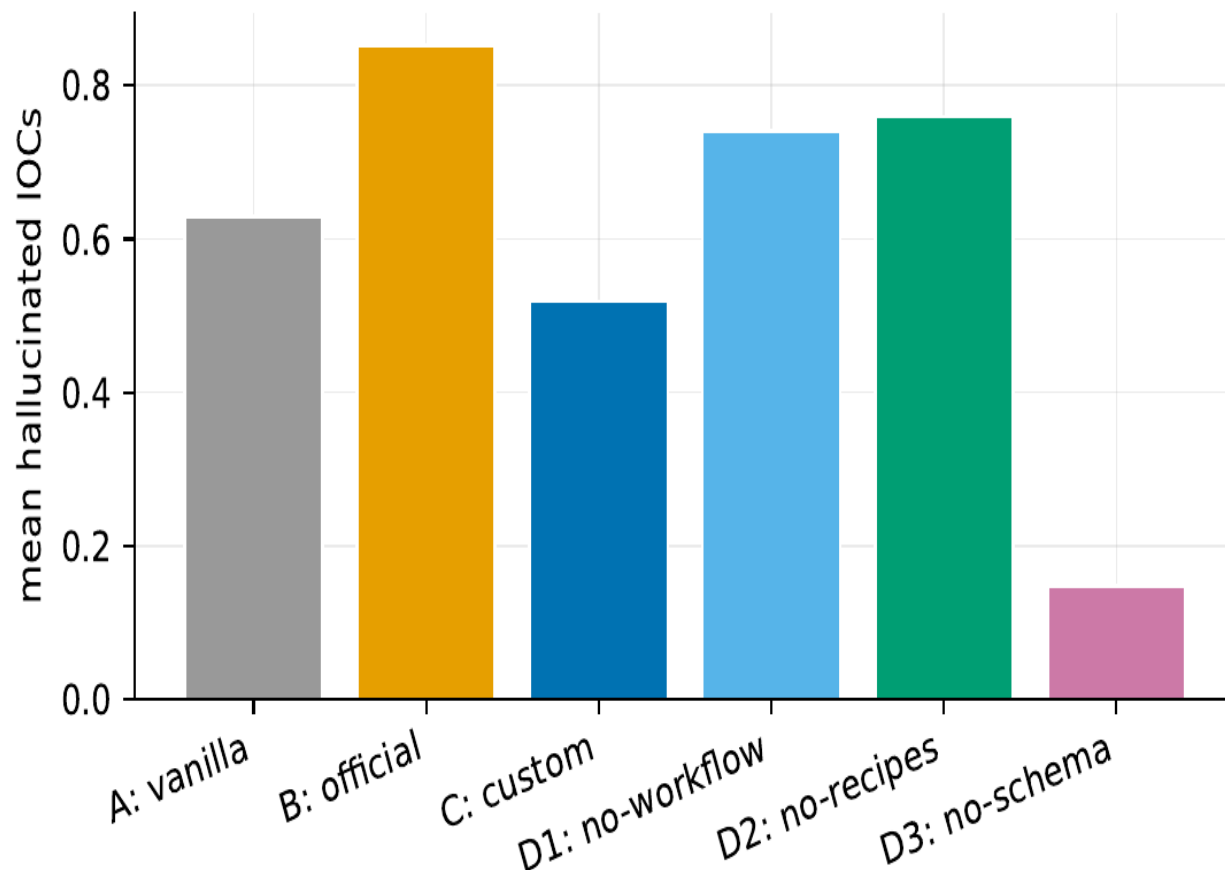
- 9 real malware-infection captures (RAT, stealer, loader families) with official analyst writeups.
- We transcribed each writeup to a typed IOC set, then checked every indicator with tshark.
- **3 of 9 answer keys printed a C2 IP absent from every packet (e.g. STRRAT key says 141.98.10.69; port-12132 traffic goes to 141.98.10.79).**
- **Corrected against primary evidence. Forensic ground truth must be validated, not trusted.**

A capable base model already identifies the victim host in 97 percent of cases, so skills barely move accuracy



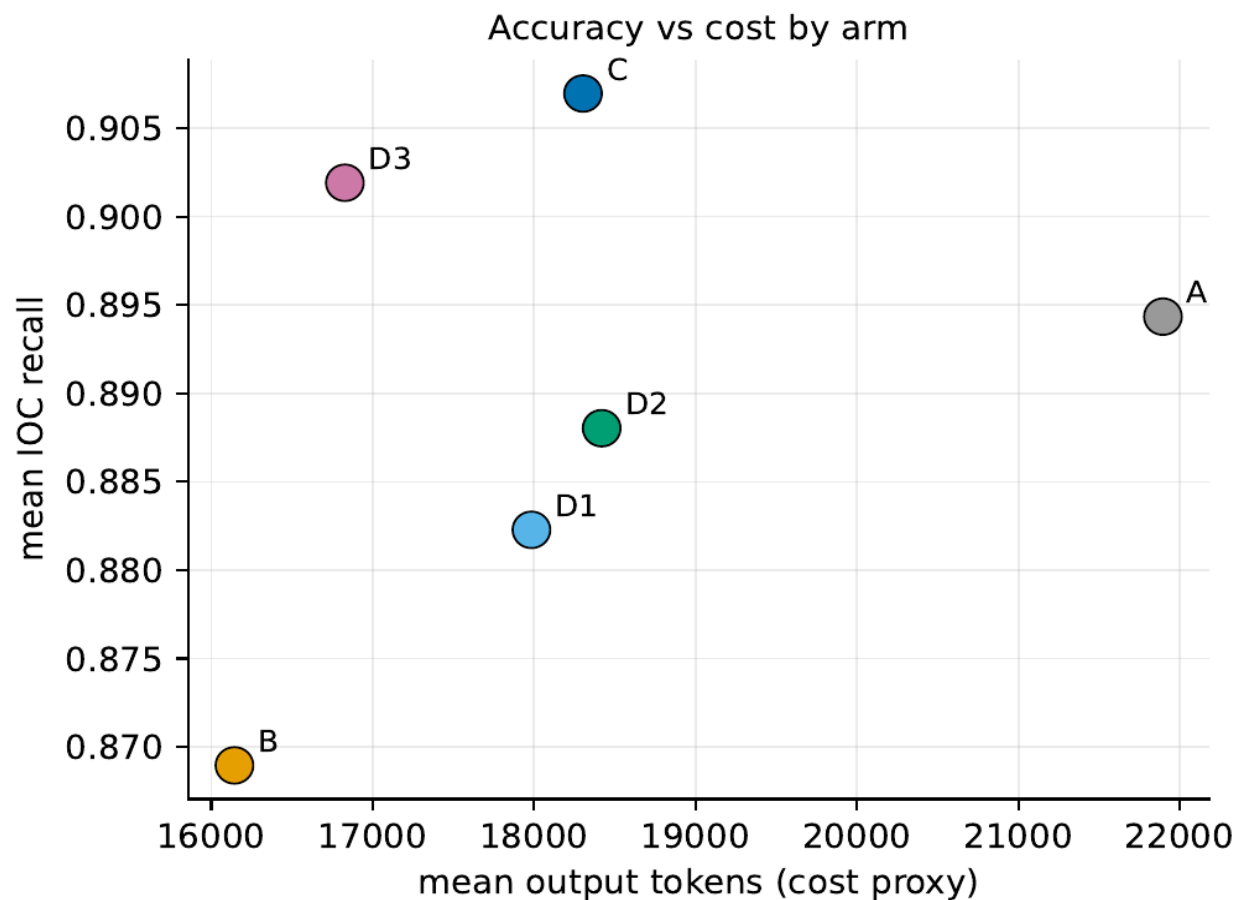
- **Victim-ID accuracy is identical (0.97) across all six arms.**
- IOC recall spans a narrow 0.87 to 0.91.
- A skill does not turn a capable model into an analyst; it already is one for routine work.
- So we look past recall: efficiency, faithfulness, and where the components matter.

A generic community skill is net-negative: it has the worst attribution and the most fabricated indicators



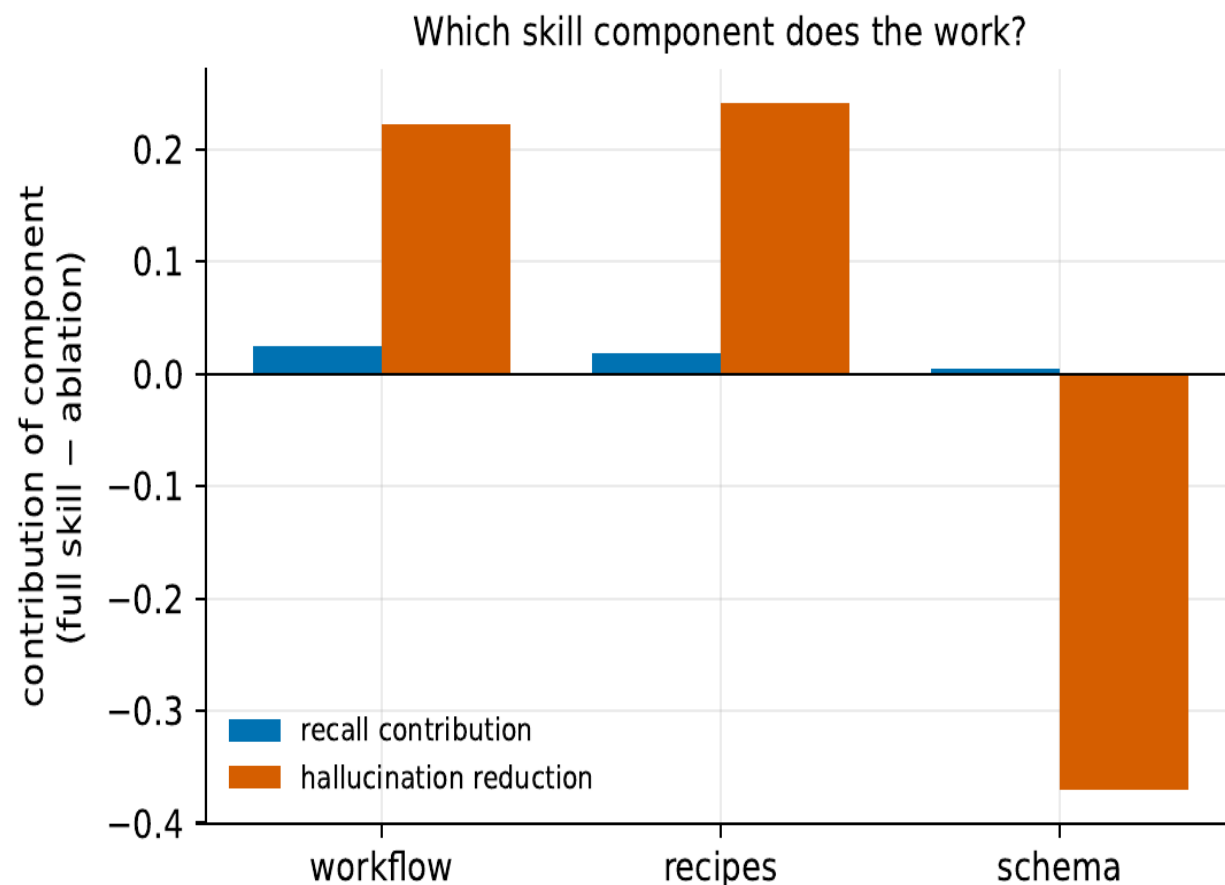
- **Official skill (B): highest hallucination (0.85 fabricated IOCs per case).**
- Also the worst family attribution (0.50 vs 0.625).
- No recall benefit over the untreated baseline.
- Injecting off-the-shelf procedural knowledge is not free.

A purpose-built skill buys efficiency and safety, not accuracy: equal recall with 37 percent fewer tshark commands



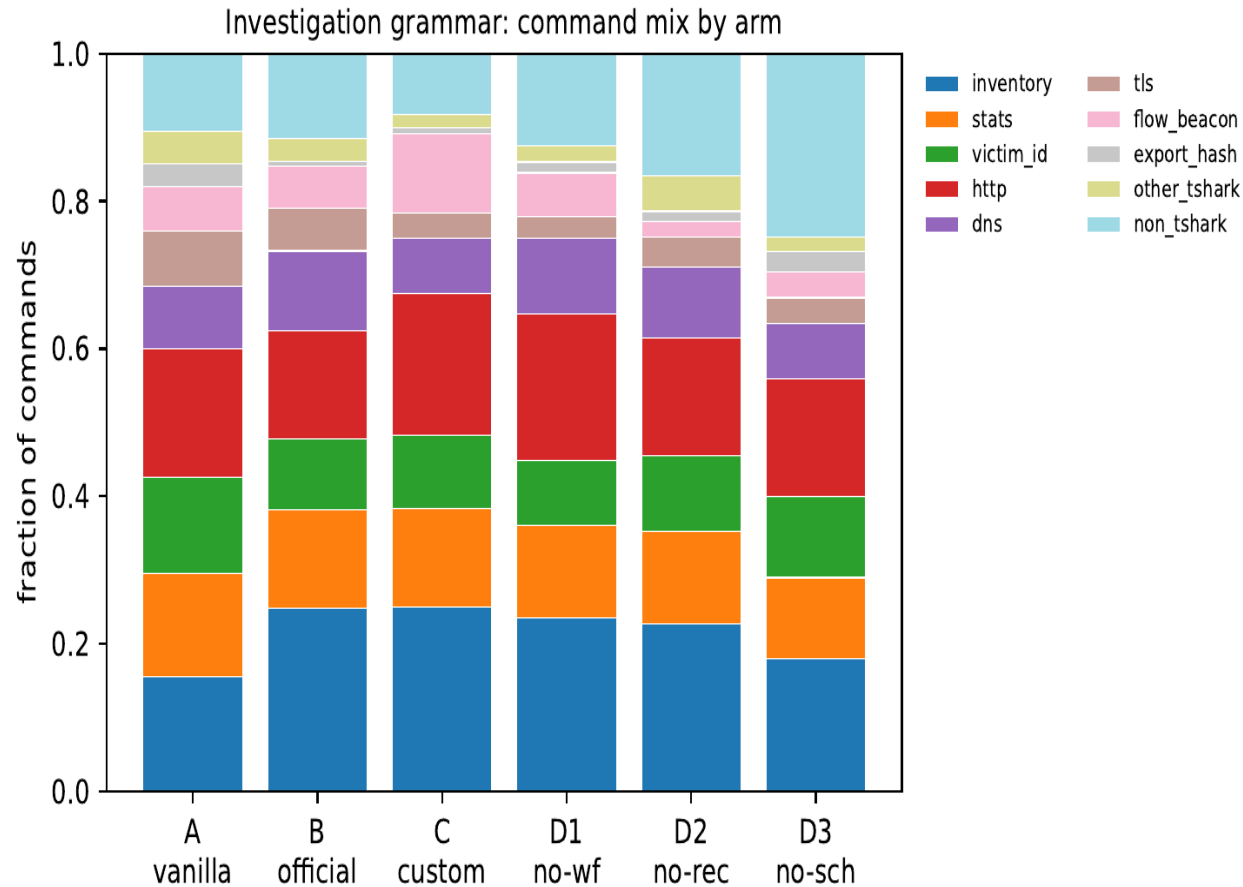
- **Custom skill (C): best recall (0.907) at the lowest command count.**
- 37% fewer tshark commands than vanilla (10.7 vs 17.0).
- Lower hallucination than both vanilla and the generic skill.
- The win is a leaner, more faithful investigation, not a higher ceiling.

Ablation shows the workflow and recipes carry the benefit, while the output schema is the largest driver of fabrication



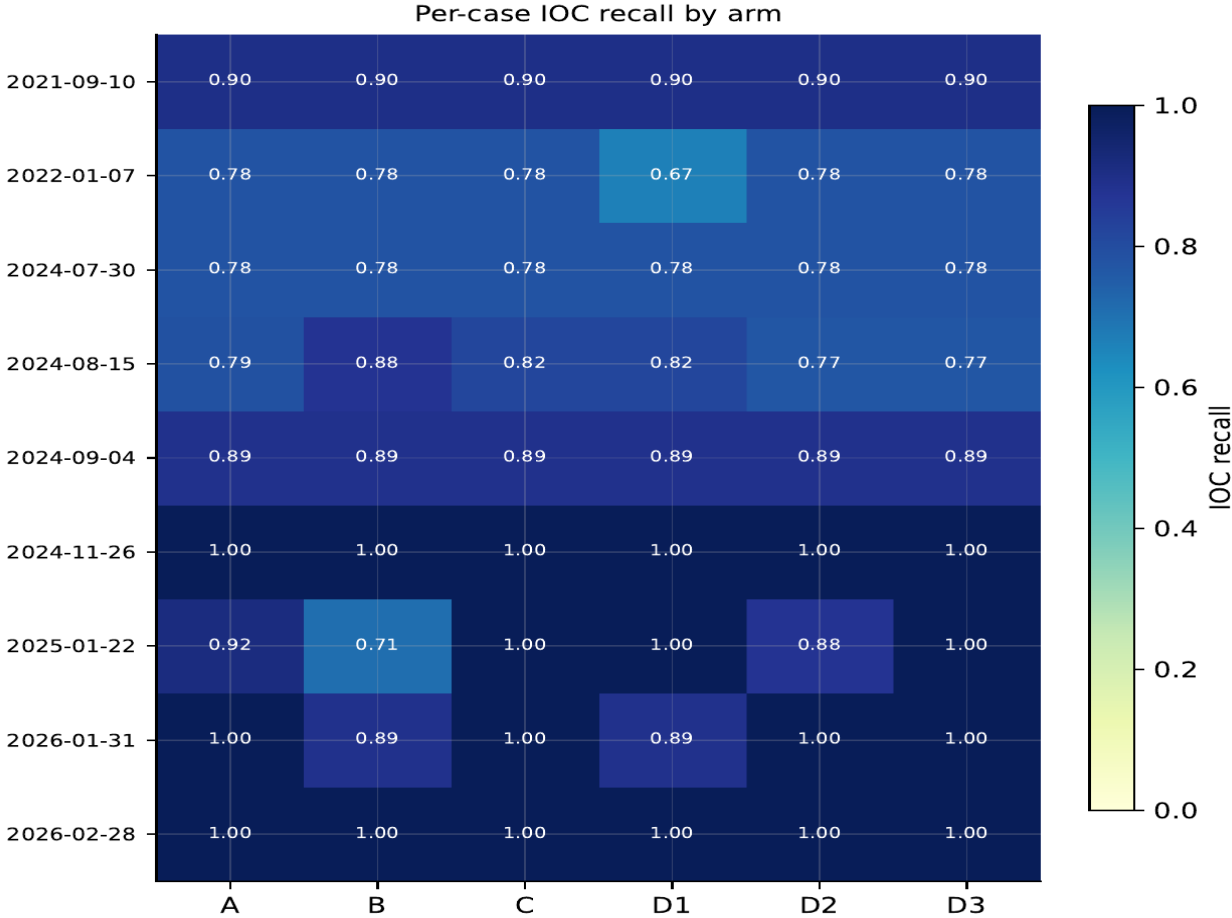
- **Removing workflow (D1): recall and attribution drop to the generic-skill level.**
- Removing recipes (D2): lowest recall, higher hallucination.
- **Removing schema (D3): recall unchanged, hallucination falls to 0.15 (best in study).**
- The rigid IOC template appears to induce fabrication: named slots invite filling.

The investigation grammar shows the custom skill runs a leaner, more targeted inquiry



- The mix of command categories each arm issues, normalized.
- The custom skill concentrates on victim-ID and statistics passes.
- It issues fewer commands overall while covering the same evidence.
- After a fix, all arms inventory the directory and read available IDS alerts.

Family attribution is the hard ceiling: no arm exceeds six of nine, bounded by alert-free behavioral cases



- Per-case IOC recall by arm; most cases sit near the ceiling.
- Where Suricata alerts are present, all arms read them and attribute correctly.
- **Where alerts are absent, no skill closes the attribution gap.**
- This bounds the promise of packet-only agentic attribution.

Rows are cases; columns are arms

Skill authors should invest in workflow and recipes, prefer evidence-gated output, and widen the evidence an agent considers

- **On a strong base model, a skill is an efficiency and safety tool, not a capability unlock.**
- Prefer evidence-gated output (emit an indicator only with a citation) over fixed templates that pressure the model to fabricate.
- Design skills that widen, not narrow, the evidence considered; a narrow skill can suppress useful exploration (the skill-issue effect).
- We release the harness, the evidence-validated dataset, and all skill variants.

References

[1] CyberSleuth: Autonomous Blue-Team LLM Agent for Web Attack Forensics. arXiv:2508.20643.[2] Holmes: Evidence-Grounded LLM Agent for Auditable DDoS Investigation. arXiv:2601.14601.[3] Before You Hand Over the Wheel: Evaluating LLMs for Security Incident Analysis. arXiv:2603.06422.[4] SkillsBench: Benchmarking How Well Agent Skills Work Across Diverse Tasks. arXiv:2602.12670.[5] Malware-Traffic-Analysis.net, B. Duncan. Traffic analysis exercises.[6] Wireshark and tshark, Wireshark Foundation, v4.6.